

```

#include <stdio.h>
#include <stdlib.h>
#define MAX 100
struct Queue {
int arr[MAX];
int front;
int rear;
};
// Function prototypes
void initializeQueue(struct Queue *queue){
queue->front =-1;
queue->rear=-1;
}
int isFull(struct Queue *queue){
return queue->rear==MAX-1;
}
int isEmpty(struct Queue *queue){
return queue->front== -1||queue->front>queue->rear;
}
void enqueue(struct Queue *queue, int element){
if(isFull(queue)){
printf("queue overflow\n");
return;}
if(queue->front== -1)
queue->front =0;
queue->rear++;
queue->arr[queue->rear]=element;
printf("%d insert into queue\n",element);
}
int dequeue(struct Queue *queue){
if(isEmpty(queue)){
printf("queue underflow\n");
return -1;
}
int element=queue->arr[queue->front];
queue->front++;
if(queue->front>queue->rear)
{
queue->front=-1;
queue->rear=-1;
}
return element;
}
void displayQueue(struct Queue *queue){
int i;
if(isEmpty(queue)){
printf("queue is empty.\n");
return ;
}
printf("queue elements: ");
for(i =queue->front;i<=queue->rear;i++){

```

```
printf("%d ",queue->arr[i]);
}
printf("\n");
}
int main() {
struct Queue queue;
int choice, element;
initializeQueue(&queue);
while (1) {
printf("\nQueue Operations Menu:\n");
printf("1. Enqueue\n");
printf("2. Dequeue\n");
printf("3. Display\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
printf("Enter element to enqueue: ");
scanf("%d", &element);
enqueue(&queue, element);
break;
case 2:
element = dequeue(&queue);
if (element != -1)
printf("Dequeued element: %d\n", element);
break;
case 3:
displayQueue(&queue);
break;
case 4:
exit(0);
default:
printf("Invalid choice! Please enter a valid option.\n");
}
}
return 0;
}
```

OUTPUT:

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter element to enqueue: 5

5 insert into queue

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter element to enqueue: 2

2 insert into queue

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter element to enqueue: 4

4 insert into queue

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 2

Dequeued element: 5

Queue Operations Menu:

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 3

queue elements: 2 4

Queue Operations Menu:

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 4

=== Code Execution Successful ===